

LendSight

End-to-End Credit Risk Analytics Platform

PROJECT REPORT

Transforming Lending Data into Actionable Portfolio Intelligence

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Power BI • SQL Server • DAX • Power Query • Excel

1. Executive Summary

LendSight is an end-to-end credit risk analytics solution designed to provide a consolidated view of lending portfolio performance, delinquency, exposure, collections, recovery and customer risk. The project combines SQL-based analysis with a Power BI reporting layer to turn lending records into actionable portfolio intelligence.

2. Business Problem

Lending organisations must monitor portfolio growth alongside credit quality. LendSight addresses the need for a centralized analytical view that allows users to move from executive portfolio health into delinquency, vintage, collections/recovery and customer-level risk analysis.

3. Objectives

Create a centralized view of portfolio health; measure exposure and credit risk; monitor DPD and PAR; evaluate collection and recovery performance; compare origination cohorts through vintage/MOB analysis; support customer-level investigation; and provide an interactive BI experience for management and analysts.

4. Data & Scope

The project uses a simulated lending portfolio with approximately 45,000 loan records and portfolio snapshot information. The analytical fields include loan and customer identifiers, disbursement and snapshot dates, DPD, loan amount, amount due, amount paid, EAD, default flag, recovery amount, status and segmentation attributes.

5. Methodology

Data preparation: prepare and transform lending records for analysis.

Risk analysis: calculate portfolio, delinquency, default, recovery and loss metrics in SQL.

BI layer: implement reusable DAX measures in Power BI.

Dashboard: present executive, delinquency, vintage, collections/recovery and customer-risk views.

Decision support: translate analytical signals into business insights and recommendations.

6. Technical Implementation

The SQL implementation covers portfolio overview, Total EAD, default rate, recovery rate, LGD, PD, ECL, PAR 30/60/90/120/180, collection effectiveness and vintage default analysis. The Power BI DAX layer includes Total Loan, Total EAD, Default Rate, Recovery Rate, Collection Effectiveness, PAR loan counts, Net Loss, Average DPD, PD and LGD.

7. Dashboard Walkthrough

Executive Overview: management-level portfolio health, exposure and credit quality.

Delinquency & DPD: repayment delinquency and severity across DPD/PAR thresholds.

Vintage Analysis: cohort performance across MOB and origination periods.

Collections & Recovery: amount due, amount paid, collection effectiveness and recovery performance.

Customer & Loan Risk Profile: segmentation and customer/loan-level investigation.

8. KPI Framework

Total Loans: loan count. **Total EAD:** default-associated exposure in the DAX implementation. **Default Rate/PD:** proportion classified as defaulted under the project methodology. **Recovery Rate:** recovery relative to relevant EAD. **LGD:** one minus recovery rate. **Collection Effectiveness:** amount paid relative to amount due. **PAR:** delinquency above defined DPD thresholds. **ECL:** $PD \times LGD \times EAD$ in the SQL implementation.

9. Key Business Insights

Portfolio growth should be assessed alongside credit quality and exposure at risk. Early delinquency represents an intervention opportunity. PAR progression provides a structured view of severity. Vintage/MOB analysis can reveal weaker origination cohorts. Recovery performance affects loss severity. Customer-level repayment and segmentation analysis can support targeted monitoring and collection strategies.

10. Strategic Recommendations

Establish recurring executive monitoring with escalation thresholds; use stage-based early collections; track migration between PAR stages; incorporate vintage performance into underwriting reviews; prioritize collection resources using exposure, delinquency severity and recovery potential; and progressively introduce predictive PD/LGD/ECL, forecasting and stress testing.

11. Limitations & Future Enhancements

The dataset is simulated and does not represent live customer performance. The analysis is primarily descriptive and diagnostic rather than a production predictive model. Monthly snapshots are less granular than daily production data. Future enhancements include automated refresh, RLS, cloud integration, predictive risk models, forecasting, stress testing and scenario analysis.

12. Conclusion

LendSight demonstrates an end-to-end workflow from lending data to credit-risk intelligence. Its value lies in connecting SQL analysis, Power BI modelling, DAX measures and business interpretation so that risk metrics become signals for portfolio monitoring, collections, underwriting review and management decision support.

Appendix — Measurement Note

The SQL PAR calculations use EAD-weighted exposure above each DPD threshold, while the Power BI PAR measures count distinct loans above each threshold. These are complementary views and should be labelled clearly in the portfolio documentation.